

Stanley Okwii

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Summary

Senior software engineer with 6+ years building production systems in Python and TypeScript on AWS, across payments, analytics products and data pipelines. Currently the founding engineer on an AI research assistant, where I own the retrieval pipeline, the agent workflow and the infrastructure that serves several client organisations from one deployment. I run AI agents through implementation, testing and review with checks at every stage, and keep architecture, security and unclear product decisions for myself. Right to work in the UK.

Skills

Programming: Python, TypeScript, JavaScript, SQL, Dart

AI/ML: RAG pipelines, LLM integration (Claude and Gemini APIs), prompt engineering, agent and tool use workflows, embeddings, vector search (pgvector), hybrid retrieval

Backend and Frameworks: Django, FastAPI, Flask, Node.js, Express, GraphQL, Celery, Redis

Frontend and Mobile: React, Next.js, Flutter

Databases: PostgreSQL, MongoDB, Redis, MySQL

Infrastructure: AWS (ECS, Bedrock, CloudFormation), GCP, Docker, GitHub Actions, Supabase, Sentry, New Relic, Terraform, Kubernetes

Testing: Unit, integration and end to end testing, Jest, Playwright

Developer Tooling: Claude Code (custom skills and agents), GitHub Copilot, Git, Antigravity CLI

Experience

Founding Engineer, Access to Energy – London, UK Jan 2026 – present

An AI research assistant that answers questions on energy access in Africa, citing evidence from a curated library of sector reports. Team of 4. I own the backend, the AI pipeline and the infrastructure.

- Built the retrieval pipeline behind the assistant, combining keyword and vector search in PostgreSQL with pgvector so questions find the right passages even when worded differently from the source.
- Designed the agent workflow that decides how to handle each question, calls the retrieval and citation tools, and declines to answer when the library holds no supporting evidence, so every answer traces back to a source.
- Built the ingestion pipeline that parses, chunks and embeds PDF reports in the background, so large uploads don't slow the app and new reports appear in search automatically.
- Designed the data model and access controls that let several research organisations share one deployment while keeping their documents fully separate and metering usage for each.
- Wrote the AWS infrastructure as CloudFormation templates and scripted environment setup, so every environment is built from reviewed code rather than manual console work.

Senior Full Stack Software Engineer, Harver – New York, USA (remote) Mar 2023 – May 2025

Promoted to Senior in March 2023. Left in May 2025 to complete my MSc full time.

- Owned the architecture and delivery of three client products (Python backend, React frontend) from design to release, making the key calls on security, performance and maintainability.
- Led the technical design of new features, working out the data model, API design and implementation plan before the team started building.

- Handled production incidents reported by clients, tracing each one through New Relic logs and traces to its root cause and shipping the fix.
- Ran planning across engineering, product and design, breaking features into milestones and raising risks early so scope could change before a deadline slipped.
- Wrote the documentation on core system workflows that new engineers used to get up to speed.
- Conducted 15+ technical interviews on system design and code quality, and helped shape how the team assessed engineering candidates.

Full Stack Software Engineer, Harver – New York, USA (remote)

Sept 2019 – Mar 2023

- Introduced structured logging and error monitoring with Sentry and New Relic, giving the team a clear trail from each production error to its root cause instead of relying on guesswork.
- Built analytics dashboards that clients used to follow talent growth and internal mobility across their organisations.
- Improved API response times by profiling the slowest endpoints, removing N+1 queries and adding caching where it mattered most.
- Automated frontend releases with CI/CD pipelines, replacing a manual deployment process with a single click.

Projects

Hoppers (Event Ticketing Platform) – Remote

Oct 2025 – present

Full Stack Software Engineer (Contract, Part Time)

- Built the payment system across several payment providers, recording commission and booking fees at the moment of purchase so revenue can be reconciled even after pricing changes.
- Implemented QR code ticketing with validation on the server, purchase limits for each buyer and safeguards that prevent the same ticket being sold twice when demand spikes.
- Delivered team management as the main contributor across the backend, admin panel and mobile app, covering invitations, role permissions and organiser dashboards.
- Built an agent workflow in Claude Code that turns a GitHub issue into a tested draft pull request, then checks its own work with Playwright screenshots for UI changes and replayed API calls for backend changes, which I verify before anything ships.
- Built a review agent that checks each pull request in an isolated dev container against our conventions, runs its own QA and follows up on replies, so feedback keeps moving while the author still has context.

CareerFoundry (Online Coding Bootcamp) – Berlin, Germany

Mar 2023 – Jan 2026

Full Stack Web Development Mentor (Freelance)

- Mentored 23+ students through the Full Stack Web Development program and held a 5/5 student rating for the quality and speed of my feedback.

Education

MSc Birkbeck, University of London, Data Analytics – London, UK

Oct 2024 – Sept 2025

- Merit. Dissertation on adversarial robustness in image classification: implemented PGD attacks from scratch and adversarially retrained ResNet50 on an ImageNet subset.

BSc Makerere University, Software Engineering – Kampala, Uganda

Aug 2014 – Sept 2018

- 2:1 (Upper Second Class Honours)